Multimodal Remote Sensing of Thunderstorm Charge Motion: A Radar Echo and Electric Field Fusion Approach

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***Abstract*—The precise observation of thunderstorm charge motion remains challenging due to the limitations of single-modal remote sensing techniques. This article proposes a novel multimodal remote sensing framework integrating radar echo intensity (REI) and atmospheric electric field (AEF) data for 3-D charge localization. Our approach tackles critical challenges in meteorological remote sensing by introducing a high-precision spatiotemporal calibration protocol to align high-frequency AEF with low-frequency radar scans, along with physics-constrained feature fusion that incorporates an AEF–REI interaction term to enhance robustness. We develop a Bayesian-optimized random forest (RF) model to improve localization accuracy under nonstationary conditions. Validated via 3-D charge trajectory reconstruction, our method achieves strong consistency with radar-observed storm core movements, demonstrating its potential as a new remote sensing paradigm for thunderstorm monitoring.**

***Index Terms*—Atmospheric electric field (AEF), Bayesian optimization, charge localization, radar, random forest (RF), remote sensing.**

I. Introduction

EXTREME weather events are occurring frequently around the world, bringing significant impacts to human production and life [1], [2]. Accurate remote sensing observation of thunderstorm charge motion is critical for early warning systems, yet existing instrumentation faces two fundamental limitations.

1) Single-modal monitoring [e.g., atmospheric electric field (AEF)] fails to capture dynamic charge–microphysics coupling [3], [4].
2) Spatiotemporal misalignment between high-frequency AEF sensors (1 s) and low-frequency radar (6 min) degrades remote sensing observation reliability [5].

Traditional remote sensing observation approaches rely on manual thresholding of AEF time-domain features (e.g., polarity reversal [6] and fast-changing jitter [7]), but struggle with the nonstationary nature of AEF signals [8]. Recent remote sensing observation studies highlight the potential of multisensor fusion for meteorological monitoring [9], [10], yet three critical gaps persist:

1) lack of standardized protocols for aligning heterogeneous sensor data (AEF signal versus radar chart);
2) inadequate physics-aware feature fusion to avoid spurious correlations;
3) sensitivity of machine learning (ML) models to hyperparameters under nonstationary conditions [11].

It is well-established that the electrical structure of a thunderstorm is typically dipolar or tripolar, with horizontally extended regions of positive charge in the upper part of the cloud and negative charge in the mid-levels, often accompanied by a smaller lower positive charge region [12], [13]. The point charge model used in this study serves as a first-order approximation to track the gross motion of the dominant charge region's effective centroid, rather than to resolve the full 3-D charge structure. This approach is justified by our goal of correlating charge motion with storm core movement observed by radar, a methodology supported by prior work that links charge layer altitudes to specific temperature ranges and radar reflectivity features [12], [14].

To address these instrumentation challenges, this article proposes a novel multisensor remote sensing observation framework integrating AEF and radar echo intensity (REI) data. Our core contributions are as follows:

1) a spatiotemporal calibration protocol resolving 1 s-versus-6-min sampling conflicts through uncertainty-aware normalization;

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2) physics-constrained feature fusion incorporating an AEF–REI interaction term to enhance remote sensing observation robustness;
3) a Bayesian-optimized random forest (RF) framework that enhances charge motion observation accuracy under nonstationary conditions, reducing the localization error compared to conventional threshold-based methods.

Validated via 3-D charge localization, our scheme achieves dynamic trajectory observation with strong consistency to radar chart observations. This work establishes a new paradigm for extreme weather instrumentation, advancing sensor fusion technology in I&M domains.

This article is organized as follows. Section II reviews existing work on AEF feature extraction for remote sensing observation, ML applications in meteorological remote sensing observation, and charge localization as an observation paradigm. Section III describes and analyzes the challenges in thunderstorm charge motion observation in this article. Section IV presents the details of the proposed remote sensing observation framework. Section V records the sources of experimental data and evaluates the performance of the proposed framework, while Section VI provides the conclusions drawn from this study.

II. Related Studies

This section summarizes and analyzes the existing work on AEF feature extraction for remote sensing observation, ML applications in meteorological remote sensing observation, and charge localization as an observation paradigm.

A. AEF Feature Extraction for Remote Sensing Observation

Existing AEF feature extraction methods focus primarily on time-domain characteristics for coarse thunderstorm assessment. Yang et al. [15] utilize preset thresholds [2-kV/m average value (AV) and 3-kV/m maximum value (MA)] to enable point charge localization, yet lack uncertainty quantification for remote sensing observation reliability. High-resolution AEF observations reveal polarity reversal dynamics during discharges [16], but fail to address spatial–temporal alignment with complementary sensors. Entropy-based weather attribute (WA) classification [5] enriches feature databases but ignores physical interpretability in classification tasks. These studies demonstrate critical gaps in I&M contexts: 1) single-modal limitations preventing holistic charge-microphysics coupling and 2) absence of standardized protocols for cross-sensor feature calibration.

B. ML Applications in Meteorological Remote Sensing Observation

While ML applications for thunderstorm-related data remain limited [17], [18], recent studies prove their efficacy in sensor fusion [19], [20], [21], [22]. RF models achieve 79% accuracy in salmon fat quantification [19], but domain-specific challenges arise: nonstationary signals and heterogeneous sensor sampling demand physics-aware architectures. For AEF classification, gradient boosting models [17], [21] define empirical prediction intervals, yet overlook hyperparameter sensitivity under observation noise. Crucially, no existing work combines Bayesian optimization with RF for adaptive observation calibration under dynamic weather conditions.

C. Charge Localization as Observation Paradigm

Charge motion visualization bridges temporal AEF observations and spatial localization. Xing et al.'s [23] foundational work derives point charge formulas with indirect error analysis, but depends on manual thresholds. Yang et al.'s [17] multipath imaging enhances precision through ML classification, yet exhibits two critical observation flaws: 1) no fusion with radar microphysics for physical validation and 2) WA evaluation gaps causing trajectory inaccuracies.

These studies uniformly neglect the critical impact of WA on observation validity in point charge motion analysis. Specifically, the absence of multimodal data-driven WA evaluation introduces errors in subsequent charge localization. This fundamentally limits their utility as quantitative meteorological remote sensing observation tools.

Furthermore, traditional point charge localization methods based on single-station E-field measurements face an inherent limitation: an infinite set of charge distributions can produce the same EF at a single point on the ground [12]. This nonuniqueness problem underscores the critical need for multisensor constraints, such as radar data, to reduce inversion ambiguity. This work addresses this by fusing REI to provide spatial context and physical validation for the motion derived from E-field inversions.

III. Challenges in Thunderstorm Charge Motion Observation

This section delineates three fundamental remote sensing observation challenges addressed by this work.

1) *Limitations of Conventional Single-Modal Remote Sensing Observation:* Existing AEF-based approaches exhibit critical shortcomings.
   a) *Physical Interpretation Gap:* Single-modal AEF data lack microphysical coupling.
   b) *Validation Deficiency:* Without radar chart cross-validation, charge motion observations cannot satisfy the uncertainty requirements.

   Our solution integrates REI data—a proven precipitation proxy in [24]—to establish a multimodal remote sensing observation framework with greater physical significance.
2) *Alignment and Fusion of Different Modal Data:* How to effectively achieve spatiotemporal alignment and feature fusion of AEF time–frequency-domain characteristics and REI data in the construction of multimodal datasets for thunderstorm charge motion is a key issue in improving the performance of classification models. This process requires addressing three primary challenges:
   a) resolving the significant temporal resolution discrepancy between the high-frequency AEF

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observations and the relatively low-frequency radar updates;

b) performing accurate spatial registration between discrete-point AEF observations and 3-D volumetric radar scan data;

c) developing an appropriate feature fusion strategy that considers the differences in physical dimensions of features, at the level of feature fusion.

3) *Adaptive Modeling for Observation Robustness:* Although RF has the ability to classify nonlinear data such as AEF, its hyperparameters are highly sensitive to features. Therefore, it is necessary to design adaptive parameter optimization strategies (such as Bayesian optimization or genetic algorithms) to dynamically balance model complexity and generalization ability.

a) On the one hand, we need to optimize the tree structure parameters (such as *max_depth*) to adapt to the local abrupt characteristics of nonstationary AEF data.

b) On the other hand, we need to adjust the feature sampling strategy (such as *max_features*) to enhance the focusing ability on key features such as the interaction term between radar echo gradient and AEF change rate.

4) *Physical Context and Limitations of Point Charge Localization:* The classical understanding of thunderstorm electrification describes a complex, multilayered charge structure [12]. Our use of a point charge model is a deliberate simplification to derive a tractable motion trajectory, not to reconstruct the full 3-D charge distribution. This approach is validated by the strong consistency between the inferred trajectory and the movement of the radar-observed storm core (Section V). However, key limitations must be acknowledged.

a) *Nonuniqueness of Inversion:* A single-point *E*-field measurement cannot uniquely determine the source charge distribution [12]. Our Bayesian-optimized RF model, trained on fused AEF-REI features, helps mitigate this by identifying patterns correlated with storm dynamics.

b) *Influence of Corona Space Charge:* When the local field exceeds ~1–3 kV/m, ground-level corona discharge can generate a layer of space charge that significantly modifies the EF measured at the sensor [25]. While the 3DAEF apparatus (3DAEFA)'s shielded, rotating-vane design mitigates errors from sensor-body coupling, it cannot fully eliminate the distorting influence of this near-ground space charge, which remains a potential source of error for single-station measurements. Our multimodal approach, which fuses AEF with REI, provides a means to identify and partially compensate for these local effects. This is achieved through the inclusion of an interaction term $I(t)$ between AEF variability and REI gradients (Section IV-C), allowing the model to learn patterns associated with precipitation and near-ground charge perturbations.

c) *Temperature-Height Context:* The altitude of charge regions is crucially linked to ambient temperature [12], [14]. Our current framework uses 2-D radar data and lacks explicit temperature profiling, which future work will integrate using 3-D radar volumes or auxiliary data.

d) *Impact of Lightning Discharges:* Lightning discharges (in-cloud and cloud-to-ground) rapidly alter the cloud's charge distribution, causing large, abrupt changes, and frequent polarity reversals in the ground-level EF [12], [13]. These discharges are a dominant source of the variability captured by features like standard deviation (SD). This introduces randomness into the instantaneous field measurements, posing a challenge for charge motion tracking. Our method addresses this by focusing on the consistent trajectory of the dominant charge region derived from the fused AEF–REI data pattern over time, rather than on interpreting instantaneous field values.

e) *Suppression of Horizontal Field by Conducting Ground:* The idealized point charge model does not account for the Earth's electrical conductivity. This omission is significant, as conductivity suppresses the horizontal EF components from cloud charges to theoretically negligible levels at the ground [12]. The measured horizontal EFs are thus primarily indicative of local, near-ground charge inhomogeneities. The model is employed herein as a tractable framework for deriving a relative motion trajectory from the vector EF data, not as a physically complete representation of the source.

f) *Simplification of Complex Charge Structure:* The point charge model is a radical simplification of the true, multilayered dipole or tripole charge structure of a thunderstorm [12], [13]. The fields from these separated charges of opposite polarity superimpose and can cancel each other, meaning an infinite set of complex charge distributions can produce the same net field at a single point on the ground. The model is not intended to represent this physical reality but to provide a tractable means of tracking the projected motion of the effective centroid of the dominant field disturbance.

IV. Proposed Remote Sensing Observation Framework

This section elaborates on the multimodal thunderstorm charge motion observation framework, which consists of five key components:

1) *AEF observation principle:* detailed working mechanism of the single-axis rotary vane structure for AEF observation (Section IV-A);

2) *multimodal data calibration:* spatiotemporal alignment protocol resolving 1 s-versus-6-min sampling conflicts, with Cohen's *d*-based feature selection (Section IV-B);

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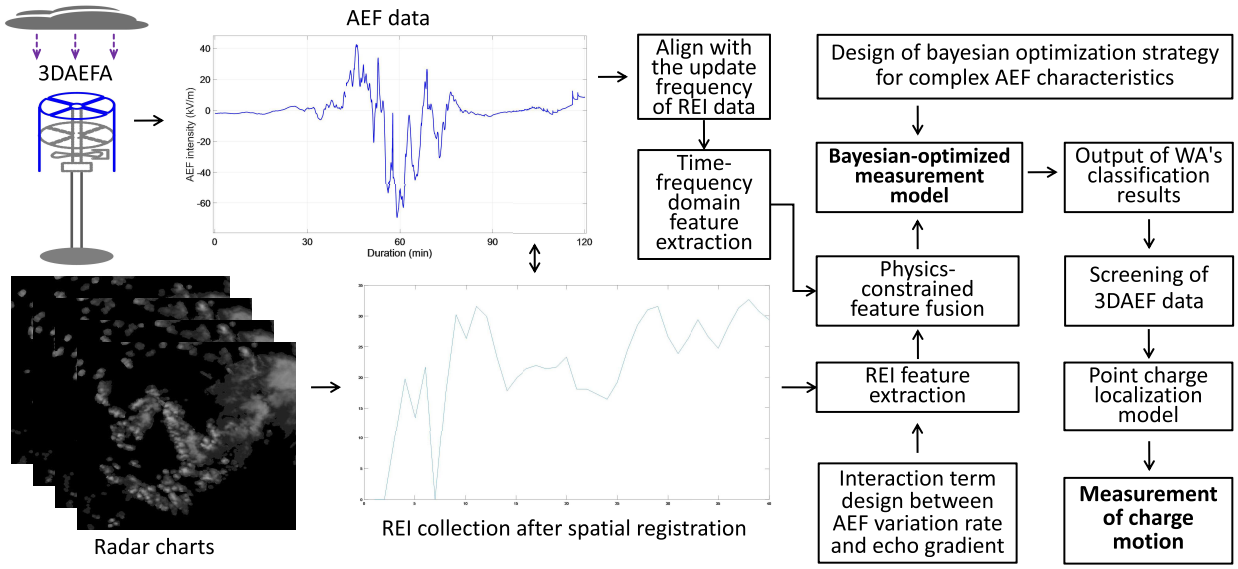


Fig. 1. Architecture of the proposed remote sensing observation framework.

3) *physics-constrained feature fusion:* normalization of heterogeneous features (SD, SE, and REI) and introduction of AEF–REI interaction term $I(t)$ (Section IV-C);
4) *Bayesian-optimized observation model:* hyperparameter metrology with Gaussian process-based uncertainty-aware optimization (Section IV-D);
5) *remote sensing observation of charge motion:* point charge localization model implementation and trajectory validation against radar charts (Section IV-E).

The complete architecture is illustrated in Fig. 1.

A. AEF Observation Principle

The AEF data used in this article are measured by 3DAEFA deployed at Nanjing University of Information Science and Technology (NUIST) station (Fig. 2).

The 3DAEFA employs a single-axis rotary vane structure based on electrostatic induction (see Fig. 3). Its remote sensing observation process involves three key mechanisms.

1) *Electrode Configuration:*
   a) Horizontal components (E_x, E_y) are measured by pairs of rectangular copper slices (X_a/X_b and Y_a/Y_b) mounted oppositely on the cylinder exterior.
   b) Vertical component (E_z) is detected via interleaved sector-shaped slices (Z_a, Z_b, $\times 4$ each) distributed azimuthally.
   c) A hollow stainless steel shielded electrode covers the induction electrodes, with fixed rings to prevent mechanical vibration during rotation.
2) *Charge Modulation:*
   a) When the shielded electrode rotates away from an induction electrode, the AEF induces opposite charges on the exposed electrode, generating a forward current.
   b) When the shield rotates over the electrode, it blocks the AEF, causing charge redistribution and a reverse current.

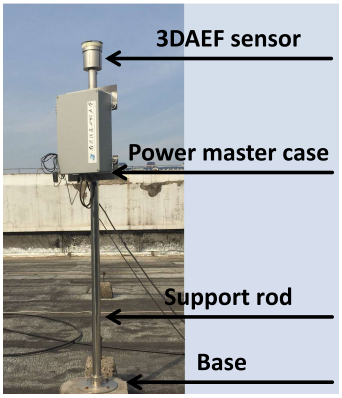


Fig. 2. 3DAEFA deployed at NUIST station.

   c) Periodic shielding yields the alternating current signals proportional to AEF components.
3) *Synchronization:*
   a) A photoelectric switch generates square-wave signals synchronized with vane rotation, providing phase References for lock-in amplification.
   b) This eliminates baseline drift and enables 3-D vector decomposition of AEF.

The rotating shielded electrode design is critical for mitigating measurement errors arising from the coupling of the dominant vertical field into the horizontal measurement components, a common challenge for grounded electrostatic field sensors. The periodic shielding and exposure mechanism prevents the persistent accumulation of charge on the sensor body, which is the primary driver of this coupling effect in a grounded configuration. Furthermore, it is acknowledged that the conducting Earth's surface imposes a boundary condition that distorts the horizontal field components from aloft, making them more vertical near the ground. Therefore, the calculated azimuth (α) and elevation (β) should be interpreted as the

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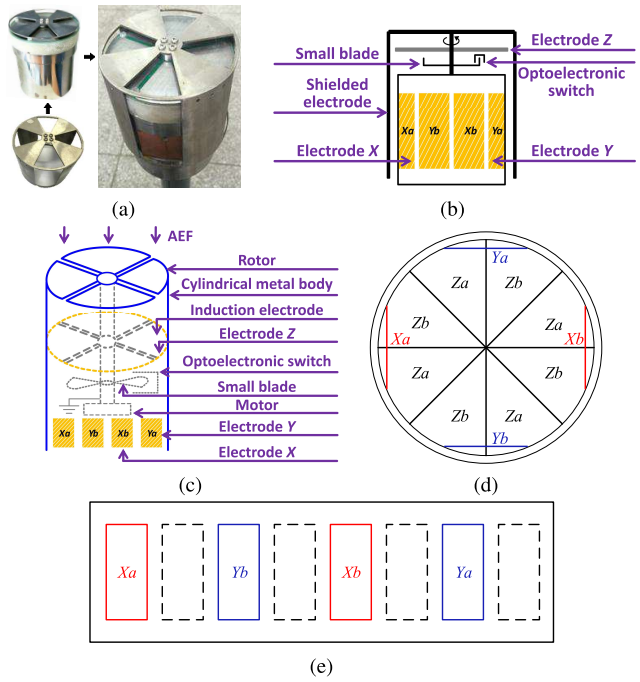


Fig. 3. Working principle and mechanical design of the 3DAEF sensor: (a) physical prototype; (b) front view showing electrode distribution; (c) internal components; (d) vertical electrode slice arrangement (top view); and (e) horizontal electrode slice layout.

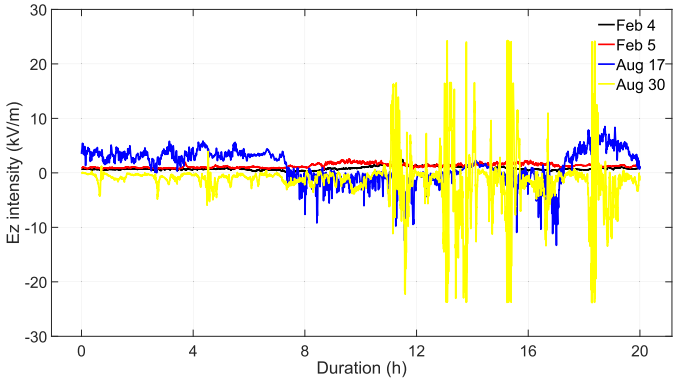


Fig. 4. E_z data samples used for AEF feature selection.

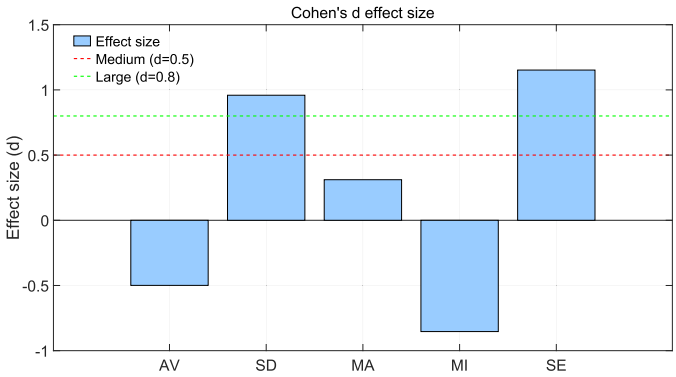


Fig. 5. Calculation results of Cohen's d.

direction to the projected source of the dominant charge disturbance for motion tracking, rather than a precise measurement of the free-air horizontal field vector.

In summary, the 3DAEFA operates as a grounded instrument, measuring the EF vector with respect to the Earth's potential. The rotating shield mechanism is the primary strategy to manage the associated challenges, such as sensor–body coupling. While this design significantly improves measurement fidelity, the inferred horizontal components (E_x and E_y) remain most representative of near-field disturbances, as discussed in Sections III and IV-E.

B. Multimodal Data Calibration

This section elaborates on the alignment and feature extraction of AEF and REI data in detail.

1) Temporal Alignment Protocol: The sampling interval T_s of AEF data is 1 s, so AEF data are represented as $x(n)$, $n = 0, 1, 2, \ldots, N-1$. Among them, N represents the total number of sampling points.

The update frequency of radar charts measured by Nanjing Radar Station is $\Delta T = 6$ min, and the number of sampling points K is 360 (i.e., $\Delta T/T_s$). Dividing $x(n)$ into an AEF component every $K = 360$, the ith AEF component $x_i(m)$ is

$$x_i(m) = x((i-1)K + m) \tag{1}$$

where $m = 0, 1, 2, \ldots, K-1$ and $i = 1, 2, 3, \ldots, N/K$. By comparing the differences in characteristic values between thunderstorm and nonthunderstorm periods, the effect quantity (Cohen's d) [26] is calculated to determine the selection of time–frequency characteristics for AEF data. Specifically, as shown in Fig. 4, vertical E_z data from 0:00 to 20:00 on February 4 and 5, 2018, are selected as nonthunderstorm samples, and E_z data from August 17 and 30, 2018, during the same time period are selected as thunderstorm samples. The Cohen's d for the AV, SD, MA, minimum value (MI), and spectral entropy (SE) features of E_z is calculated, as shown in Fig. 5.

In Fig. 5, Cohen's d values of 0.5 and 0.8 are used as the medium (red dashed line) and large effect (green dashed line) intensity thresholds, respectively. If the Cohen's d value obtained from the calculation of a certain feature is not less than 0.8, it means that there is a significant difference in feature performance between thunderstorm and nonthunderstorm samples, with clear discrimination. Therefore, SD ($d = 0.96$) and SE ($d = 1.15$) are selected as the key features for AEF data. These features are effective as they encapsulate the dynamics and complexity of the AEF signal.

2) Spatial Registration and REI Feature Extraction: The radar charts [27], [28] measured by Nanjing Radar Station cover the entire Jiangsu Province of China, with an east longitude λ and north latitude ϕ range of

$$\begin{cases} \lambda \in [\lambda_{\min}, \lambda_{\max}] = [116.35^\circ, 121.93^\circ] \\ \phi \in [\phi_{\min}, \phi_{\max}] = [30.75^\circ, 35.13^\circ]. \end{cases} \tag{2}$$

When dividing the corresponding radar chart area into a uniform 560 × 480 pixel grid, the resulting resolutions in

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longitude and latitude directions are $\Delta\lambda$ and $\Delta\phi$, respectively,

$$\begin{cases} \Delta\lambda = \dfrac{\lambda_{\max} - \lambda_{\min}}{560} = \dfrac{121.93^\circ - 116.35^\circ}{560} \\ \qquad\quad \approx 0.00996^\circ/\text{pixel} \\ \Delta\phi = \dfrac{\phi_{\max} - \phi_{\min}}{480} = \dfrac{35.13^\circ - 30.75^\circ}{480} \\ \qquad\quad \approx 0.00913^\circ/\text{pixel}. \end{cases} \tag{3}$$

At the latitude of the NUIST station ($\phi_{\text{NUIST}} = 32.2066^\circ$), one degree of longitude corresponds to approximately 94.2 km, and one degree of latitude corresponds to approximately 111 km. This results in an approximate spatial resolution of 0.94 km/pixel in the longitudinal direction and 1.01 km/pixel in the latitudinal direction.

The radar chart data $\mathbf{R}$ are a 480×560 matrix

$$\mathbf{R} = \begin{bmatrix} r_{1,1} & r_{1,2} & \cdots & r_{1,560} \\ r_{2,1} & r_{2,2} & \cdots & r_{2,560} \\ \vdots & \vdots & \ddots & \vdots \\ r_{480,1} & r_{480,2} & \cdots & r_{480,560} \end{bmatrix}, \quad r_{i,j} \in \mathbb{R}. \tag{4}$$

Among them, $r_{i,j}$ represents the REI of the ith row and jth column pixels. The latitude and longitude of NUIST station are $(\lambda_{\text{NUIST}}, \phi_{\text{NUIST}}) = (118.7172^\circ, 32.2066^\circ)$, and its corresponding grid row and column index $(i_{\text{NUIST}}, j_{\text{NUIST}})$ are

$$\begin{cases} j_{\text{NUIST}} = \left\lfloor \dfrac{\lambda_{\text{NUIST}} - \lambda_{\min}}{\Delta\lambda} \right\rfloor + 1 = 240 \\ i_{\text{NUIST}} = \left\lfloor \dfrac{\phi_{\text{NUIST}} - \phi_{\min}}{\Delta\phi} \right\rfloor + 1 = 153. \end{cases} \tag{5}$$

Therefore, at time t, the REI(t) data corresponding to the NUIST station are $r_{153,240}(t)$, where t represents the AEF data collection time. As described in the challenges in Section III, it is necessary to spatially register AEF observations at discrete points with REI data. For this purpose, the REI value $r_{153,240}(t)$ of the pixel where the NUIST station is located is directly extracted as the radar chart feature.

While the radar provides 3-D volumetric scans, the REI feature $r_{153,240}(t)$ extracted for fusion corresponds to the lowest elevation angle (0.5°)-plan position indicator data. This simplification is made for initial spatial registration with the single-point AEF sensor. We acknowledge that a more physically complete analysis would utilize the 3-D radar volume to correlate reflectivity with charge layers at specific temperature altitudes (e.g., −10 °C to −20 °C level where the main negative charge is often located [12], [14]). This represents a key direction for future work.

C. Physics-Constrained Feature Fusion

Considering that the physical dimensions of the aligned AEF features SD [denoted as SD(t)], SE [denoted as SE(t)], and REI(t) are different, they are normalized

$$\begin{cases} \overline{\text{SD}}(t) = \dfrac{\text{SD}(t) - \min(\text{SD})}{\max(\text{SD}) - \min(\text{SD})} \\ \overline{\text{SE}}(t) = \dfrac{\text{SE}(t) - \min(\text{SE})}{\max(\text{SE}) - \min(\text{SE})} \\ \overline{\text{REI}}(t) = \dfrac{\text{REI}(t) - \min(\text{REI})}{\max(\text{REI}) - \min(\text{REI})}. \end{cases} \tag{6}$$

Among them, $\min(\cdot)$ and $\max(\cdot)$, respectively, represent the minimum and maximum values of the feature in the target dataset. The normalized $\overline{\text{SD}}(t)$, $\overline{\text{SE}}(t)$, and $\overline{\text{REI}}(t)$ all fall within $[0, 1]$. Here, using the concatenation method suitable for RF model, the three are directly concatenated into a 3-D feature vector $\mathbf{F}_{\text{fused}}(t)$

$$\mathbf{F}_{\text{fused}}(t) = \left[\overline{\text{SD}}(t), \overline{\text{SE}}(t), \overline{\text{REI}}(t)\right] \in \mathbb{R}^3. \tag{7}$$

In order to introduce the interaction term between AEF change rate (AEF change gradient) and radar echo gradient (REI spatial gradient), the change rate $\Delta\text{SD}(t)$ in the time dimension is first calculated

$$\Delta\text{SD}(t) = \frac{\text{SD}(t+1) - \text{SD}(t-1)}{2}. \tag{8}$$

Radar data $\mathbf{R}(t)$ are a 2-D image with row and column indices i and j, from which the spatial gradient $\nabla\text{REI}(t)$ of REI is calculated

$$\nabla\text{REI}(t) = \sqrt{\left(\frac{\partial\text{REI}(t)}{\partial i}\right)^2 + \left(\frac{\partial\text{REI}(t)}{\partial j}\right)^2}. \tag{9}$$

Define interaction term $I(t)$ as the product of normalized AEF change rate $\Delta\overline{\text{SD}}(t)$ and spatial gradient $\nabla\overline{\text{REI}}(t)$

$$I(t) = \Delta\overline{\text{SD}}(t) \times \nabla\overline{\text{REI}}(t). \tag{10}$$

This interaction term $I(t)$ is designed to capture coupled behavior between AEF dynamics and precipitation-related radar echoes, which may include signals from near-ground charge carriers (e.g., charged precipitation or corona space charge). By incorporating this term, the model can learn to distinguish between field variations caused by large-scale cloud charge motion and those influenced by local, near-ground phenomena.

Combine interaction item $I(t)$ with the original concatenated feature $\mathbf{F}_{\text{fused}}(t)$ to form the final fused feature $\mathbf{F}_{\text{final}}(t)$

$$\mathbf{F}_{\text{final}}(t) = \left[\overline{\text{SD}}(t), \overline{\text{SE}}(t), \overline{\text{REI}}(t), I(t)\right] \in \mathbb{R}^4. \tag{11}$$

D. Bayesian Optimization of RF Classifier

1) Hyperparameter Metrology: Based on the feature matrix $\mathbf{F}_{\text{final}}(t)$ and label vector $\mathbf{L}(t)$, Bayesian optimization [29], [30] is used to find the optimal hyperparameter combination θ^* (including the number of trees N_{trees}, maximum depth $D_{\max}$, and feature subset size $F_{\max}$) for the RF, in order to minimize the classification error rate $\min_{\theta\in\Theta} \mathcal{L}(\theta)$ of fivefold cross-validation

$$\min_{\theta\in\Theta} \mathcal{L}(\theta) = \frac{1}{K}\sum_{k=1}^{K}\left(\widehat{\mathbf{L}}_k(\theta) \neq \mathbf{L}_k\right). \tag{12}$$

Among them, $K = 5$ is the cross-validation fold. The hyperparameter combination is

$$\theta = (N_{\text{trees}}, D_{\max}, F_{\max})$$
$$\Theta = \begin{Bmatrix} N_{\text{trees}} \in [10, 200] \cap \mathbb{Z}^+ \\ D_{\max} \in [1, 20] \cap \mathbb{Z}^+ \\ F_{\max} \in [1, \min(4, n_{\text{features}})] \cap \mathbb{Z}^+ \end{Bmatrix}. \tag{13}$$

Among them, n_{features} is the actual number of columns in the feature matrix $F_{\max}(t)$. This dynamic interval ensures that the optimizer always searches within the valid parameter space, avoiding running errors.

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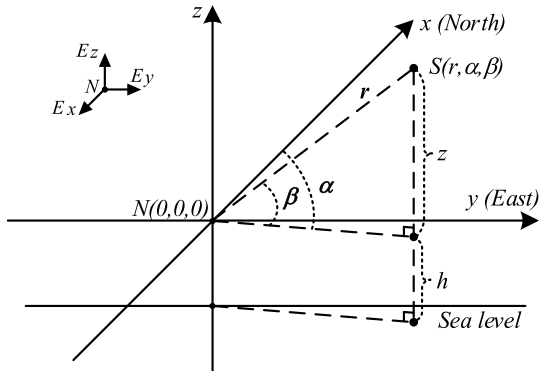


Fig. 6. Point charge localization model based on 3DAEFA.

2) Uncertainty-Aware Optimization: Modeling the objective function through a Gaussian process

$$\mathcal{L}(\theta)\widetilde{\mathcal{GP}}\left(\mu(\theta), k\left(\theta,\theta'\right)\right). \tag{14}$$

Select the next evaluation point through iteration

$$\begin{cases}\theta_{n+1} = \arg\max\limits_{\theta\in\Theta} \alpha_{\mathrm{EI}}(\theta), \\ \alpha_{\mathrm{EI}}(\theta) = \left[\max\left(0, \mathcal{L}^* - \mathcal{L}(\theta)\right)\right].\end{cases} \tag{15}$$

3) Model Implementation: The optimal RF model is

$$f^*\left(\mathbf{F}_{\mathrm{final}}\right) = \mathrm{mode}\left(\left\{T_i\left(\mathbf{F}_{\mathrm{final}}; D^*_{\max}, F^*_{\max}\right)\right\}_{i=1}^{N^*_{\mathrm{trees}}}\right). \tag{16}$$

The splitting rule SplitNode for a single tree is

$$\begin{aligned}\mathrm{SplitNode} &= \arg\max_{j\in\mathcal{S}}(\mathrm{Gain}(j)) \\ \mathcal{S} &\subset \{1,2,3,4\}, \quad |\mathcal{S}| = F^*_{\max}.\end{aligned} \tag{17}$$

In the Bayesian optimization process, first, five sets of parameters are randomly initialized in the preset hyperparameter space (N_{trees} is between 10 and 200, $D_{\max}$ is between 1 and 20, and $F_{\max}$ is not less than 1), followed by 30 rounds of iterative optimization: a Gaussian process proxy model is constructed based on historical evaluation data in each round, and the next candidate parameter combination is intelligently selected through the expected improvement function. The performance is evaluated using fivefold cross-validation, and the optimal parameter combination θ^* that minimizes the validation error is finally output. Based on θ^*-trained RF model, each decision tree captures the electric radar coupling characteristics by selecting split points from $F^*_{\max}$ random features [including the interaction term $I(t)$] within the depth limit of $D^*_{\max}$, and finally integrates N^*_{trees} trees for output. While ensuring the ability to model complex large gas electrical features, the overfitting risk is effectively controlled through feature subset sampling and depth constraints.

E. Observation of Charge Motion

Fig. 6 shows the point charge localization model for thunderstorm clouds [3], [15].

In Fig. 6, 3DAEFA is located at $N(0,0,0)$, with an altitude of h. The Cartesian and spherical coordinates of point charge S are $S(x,y,z)$ and $S(r,\alpha,\beta)$, respectively. The calculation formula for $S(r,\alpha,\beta)$ is

$$r = \sqrt[4]{\frac{A^2(1-B)^2}{{E_x}^2 + {E_y}^2 + {E_z}^2}} \tag{18}$$

$$\alpha = \begin{cases}\arctan\dfrac{E_y}{E_x}, & \text{when } E_x > 0 \\ \arctan\dfrac{E_y}{E_x} + 180°, & \text{when } E_x < 0 \text{ and } E_y > 0 \\ \arctan\dfrac{E_y}{E_x} - 180°, & \text{when } E_x < 0 \text{ and } E_y < 0\end{cases} \tag{19}$$

$$\beta = \arctan\frac{\varepsilon_1 |E_z|}{\varepsilon_2\sqrt{{E_x}^2 + {E_y}^2}}. \tag{20}$$

In (18), $A = -(q/4\pi\varepsilon_1)$ and $B = (\varepsilon_2 - \varepsilon_1)/(\varepsilon_2 + \varepsilon_1)$. ε_1 and ε_2, respectively, represent the permittivities of the air and ground, where 3DAEFA is located. E_x and E_y represent the horizontal AEF components in the x- and y-directions, respectively, while E_z represents the vertical AEF component. The steps for measuring the motion of point charges are as follows.

1) AEF and REI data are used to obtain multimodal data feature vectors with physical interaction terms.
2) The feature vector and label vector are input into the optimal RF model to obtain WA classification results.
3) The 3DAEF data corresponding to the thunderstorm dominant attribute classified by the RF model are substituted into (18)–(20) to obtain the point charge localization results.
4) The localization results are fit in the form of polynomial curves from the *X0Y* plane perspective [17], [31] to measure and visualize the trajectories of point charge motion. These trajectories are compared with the movement trend of the thunderstorm core area in the radar charts to verify the effectiveness of the proposed method.

This study employs the point charge model as a tractable framework to derive a quantifiable motion trajectory from the temporal evolution of the measured 3DAEF vector. However, a critical consideration in interpreting the results (e.g., horizontal angle α and elevation angle β) is the electrical conductivity of the Earth. The model presented assumes a dielectric half-space, whereas for quasi-static fields, the ground's conductivity dominates its complex permittivity, drastically suppressing the horizontal field component originating from cloud charges [12], [13]. Consequently, the significant horizontal fields measured by the 3DAEFA cannot be primarily attributed to cloud charges via this idealized model. They are instead interpreted as being generated by local phenomena, most notably horizontal gradients in space charge density from corona discharge and charged precipitation advected by wind. Therefore, the trajectory derived herein should not be interpreted as the precise path of a cloud-based point charge, nor does it resolve the complex multipolar charge structure. It provides a consistent projection of the motion of the centroid of the dominant EF disturbance, the validity of which is empirically supported by its strong correlation with the movement of the radar-observed storm core, as demonstrated in Section V.

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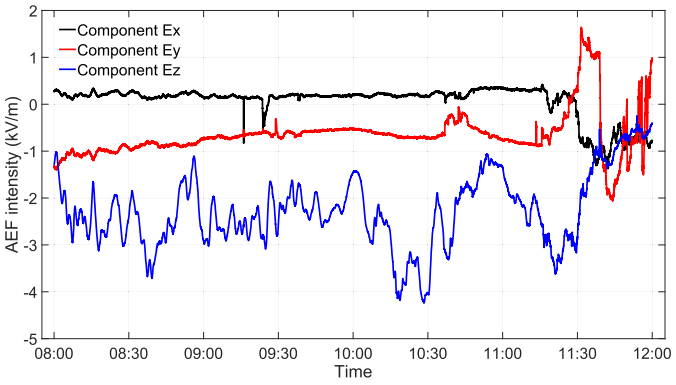


Fig. 7. 3DAEF data from 8:00 to 12:00 on September 3, 2019.

V. Experimental Results and Analyses

This section presents experimental validation of the proposed framework, including: 1) preparation of calibrated multimodal data from 8:00 to 12:00 on September 3, 2019; 2) performance benchmarking against threshold-based and entropy-based methods in temporal alignment and spatial localization; 3) evaluation of remote sensing observation models with comparisons between Bayesian-optimized and nonoptimized RF parameters and other classifiers; and 4) analysis of point charge motion trajectories validated against radar observations. All data are recorded in Beijing time.

A. Calibrated Multimodal Dataset

The 3DAEF (Fig. 7) and radar chart (Fig. 8) data from 8:00 to 12:00 on September 3, 2019, are selected. The selected case from September 3, 2019, represents the passage of a large-scale convective system. This presents a more complex scenario than an isolated thunderstorm but serves to test the framework's capability in a realistic, operationally relevant context. The first row of charts in Fig. 8 corresponds from 8:00 to 8:24 from left to right, the second row of charts corresponds from 8:30 to 8:54 from left to right, and so on. The normalized feature values SD, SE, and REI are plotted against time (in 6-min intervals) from 8:00 to 12:00 on September 3, 2019, as shown in Fig. 9.

In Fig. 7, AEF components change dramatically, especially the horizontal component E_y and the vertical component E_z. Based on experience, it is highly likely that thunderstorm activity occurs above NUIST during this period. The numerical changes of features SD, SE, and REI shown in Fig. 9 reflect the strong volatility of these main features and the presence of severe weather at the data level.

B. Observation Performance Benchmarking

The point charge localization performance is compared across the following.

1) *Threshold-Based Method [15]:* Conventional AEF amplitude threshold, i.e., MA.
2) *Entropy-Based Method [5]:* The weather conditions are distinguished through entropy value intervals to provide WA information for point charge localization.

TABLE I
Performance Comparison of Remote Sensing Observation Methods

Method	R_{time} (%)	RMSE (km)
Threshold-based [15]	53	1.32
Entropy-based [5]	74.2	0.81
Proposed	92.4	0.56

TABLE II
Performance Comparison of Models Under Different Parameter Settings

No.	N_{trees}	$D_{\max}$	$F_{\max}$	Acc.	Pre.	F1
1	186	17	15	0.833	0.778	0.875
2	41	18	8	0.75	0.75	0.8
3	151	14	9	0.833	0.857	0.857
4	91	14	5	0.667	0.714	0.714
5	175	3	1	0.75	0.7	0.824
6	161	7	16	0.917	0.875	0.933

3) *Proposed Framework:* Full framework with Bayesian-optimized RF.

The remote sensing observation performance is evaluated by comparing thunderstorm period detection accuracy and charge localization precision against radar observations. The radar ground truth is defined by echo intensity > 35 dBZ at the lowest tilt, which serves as a proxy for the location and movement of the strong convective core where charge separation is most likely active [12]. It is acknowledged that this reflectivity-based ground truth does not uniquely identify electrified regions and that the measured AEF may include contributions from charged precipitation near the sensor. Temporal alignment is quantified through the overlap ratio $R_{\text{time}} = |T_{\text{method}} \cap T_{\text{truth}}|/|T_{\text{truth}}|$ between detected periods T_{method} and true thunderstorm periods T_{truth}, while spatial accuracy is measured via root-mean-square error (RMSE) = $\left((1/N)\sum_{i=1}^{N} \|\mathbf{p}_i - \mathbf{p}_{\text{radar}}\|^2\right)^{1/2}$ of localized charges relative to radar core centroids.

In Table I, the proposed framework demonstrates superior performance with 92.4% temporal alignment accuracy, outperforming the entropy-based method (74.2%) and threshold-based method (53%). In spatial localization, our method achieves 0.56-km precision, representing 30.9% and 57.6% improvements over the entropy-based (0.81 km) and threshold-based (1.32 km) methods, respectively. These results confirm that multimodal sensor fusion combined with Bayesian optimization provides the most accurate thunderstorm charge motion remote sensing observation among the compared techniques.

C. Comparative Performance of Classification Models

1) Comparison of Parameter Settings: Five sets of hyperparameter groups (numbered 1–5) for RF models (without Bayesian optimization) are randomly generated. The data prepared in Section V-A are used to compare the performance of the RF model with unoptimized parameters and the optimized model (No. 6) we established. Accuracy (Accu.), Precision (Pre.), and $F1$ score ($F1$) are selected as performance evaluation indicators, and the results are shown in Table II.

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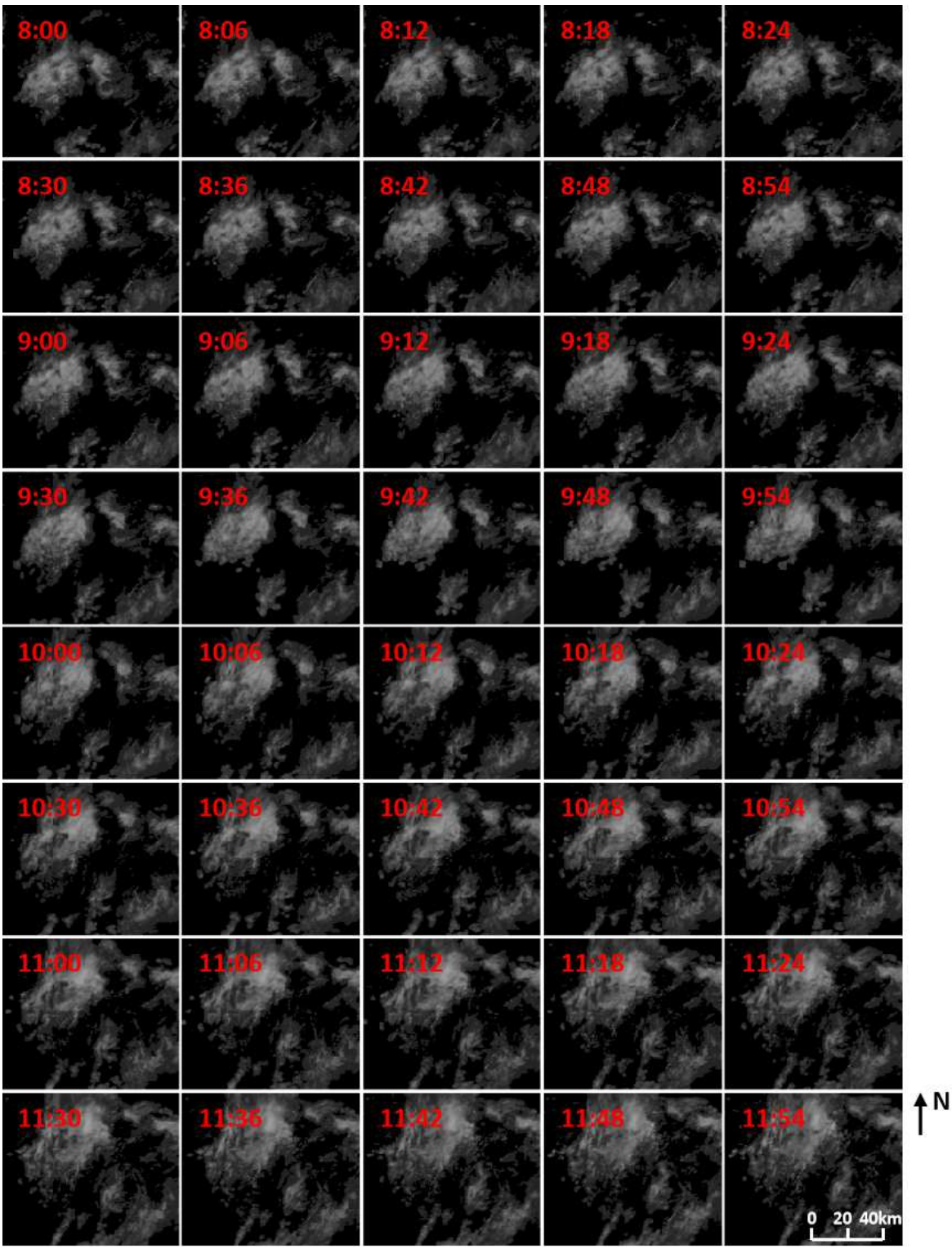


Fig. 8. Radar charts from 8:00 to 12:00 on September 3, 2019.

In Table II, the established optimization model is the most competitive in all indicators, with accuracy, precision, and $F1$ score of 0.917, 0.875, and 0.933, respectively. The values of N_{trees}, $D_{\max}$, and $F_{\max}$ contributed by Bayesian optimization are 161, 7, and 16, respectively. Compared to the unoptimized parameter model, the optimized model achieves a maximum accuracy, precision, and $F1$ score improvement of 37.5%, 25%, and 30.7%, respectively. Even the smallest improvements have reached 10.1%, 2.1%, and 6.6%. This is sufficient to demonstrate the effectiveness of the proposed model optimization scheme.

2) Comparison of Model's Performance: Based on Fig. 9, the multimodal feature data from 8:00 to 11:00 (Sample No.1) and 8:00 to 12:00 (Sample No.2) on September 3, 2019, are

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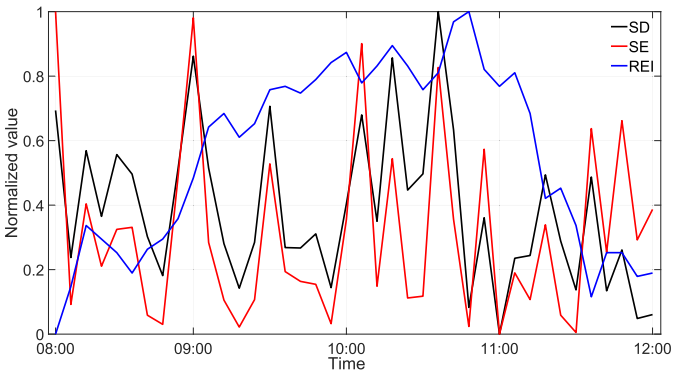


Fig. 9. Temporal variations of normalized feature values (SD, SE, and REI) from 8:00 to 12:00 on September 3, 2019.

TABLE III
PERFORMANCE COMPARISON RESULTS OF CLASSIFICATION MODELS

Model	Sample No.1			Sample No.2		
	Acc.	Pre.	F1	Acc.	Pre.	F1
SVM	0.75	0.727	0.842	0.833	0.778	0.875
LM	0.667	0.667	0.8	0.75	0.7	0.8
Our model	0.875	0.8	0.889	0.917	0.875	0.933

used as two samples to compare the performance of different classification models. Conventional parameter settings are applied to both the support vector machine (SVM) and logit model (LM). The performance evaluation results are shown in Table III.

In Table III, our model performs the best under different samples. In the first sample test, the proposed model achieves the accuracy, precision, and $F1$ score improvements of 16.7%, 10%, and 5.6%, respectively, compared to the suboptimal SVM. In the second sample test, the proposed model achieves the accuracy, precision, and $F1$ score improvements of 10.1%, 12.5%, and 6.6%, respectively, compared to the suboptimal SVM. The proposed model achieves more improvement compared to LM. This is mainly because LM has relatively weak nonlinear processing capabilities, making it difficult to directly apply to complex AEF data feature processing [32]. The superior performance demonstrated by the proposed model provides a WA label filtering guarantee for the subsequent visualization of point charge motion.

D. Observation Results of Charge Motion

According to the classification results obtained using the proposed framework, 3DAEF data from 8:54 to 11:12 carry thunderstorm dominant attributes. By substituting these data into (18)–(20), the point charge localization and motion trajectory on the *X0Y* plane (black curve) as shown in Fig. 10 are obtained.

In Fig. 10(a), the development of time corresponds to the color change of point charges from red to green. A large amount of point charges have accumulated near the origin (i.e., 3DAEFA), indicating intense thunderstorm activity. In Fig. 10(b), the main motion of point charges is in the North–South direction. They originate in the southwest direction and move northward over time (indicated by the black

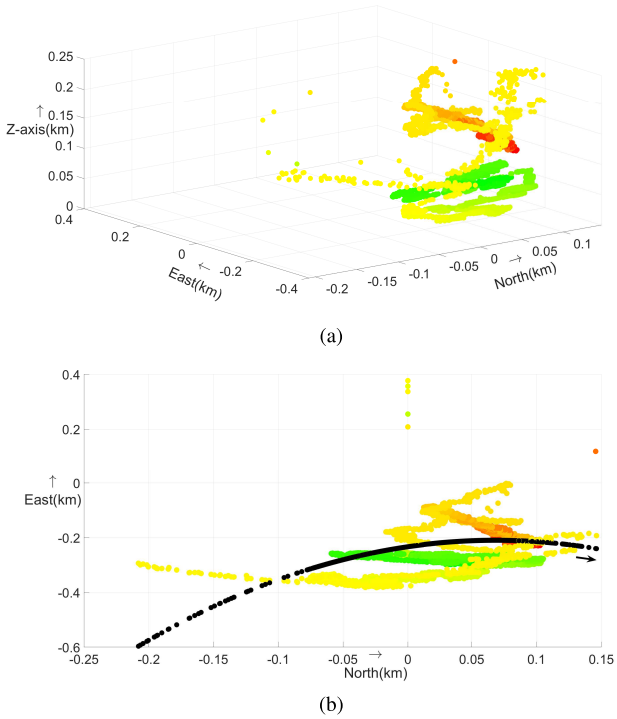


Fig. 10. Results of point charge (a) localization and (b) motion trajectory.

arrow). In Fig. 8, the pixel point (153, 240) is located at the same position as the NUIST station. From this perspective, the movement of the core thunderstorm area in the radar charts is observed, and the thunderstorm area is moving northward over time. These observation results are consistent with the analyses of point charge motion, strongly proving the effectiveness of the proposed method.

It is noted that the point charge trajectories derived from ground-based E-field measurements are influenced by near-ground physical processes (e.g., corona space charge and precipitation charge) and by lightning discharges which abruptly redistribute charge aloft. An inherent interpretational consideration is that the inferred trajectory could potentially represent the sequential influence of different, localized charged regions (e.g., embedded convection) rather than the smooth motion of a single dominant charge center. However, the strong, consistent directional trend of the derived trajectory and its close correlation with the northward movement of the entire radar-observed storm core (Fig. 8) provide compelling evidence that our method is tracking the projected gross motion of the storm's electrified system, rather than unconnected local events. While the absolute accuracy of the inferred charge location is subject to these perturbations [33], the consistency in motion tracking remains valid.

Future work involving multistation measurements, 3-D radar data, and lightning mapping arrays will help to further constrain these effects. A key next step will be to apply this framework to well-observed cases of isolated cumulus development, which provide a simpler and more interpretable meteorological context for validating the charge motion tracking capability.

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VI. Discussion and Conclusion

This article proposes a novel framework for thunderstorm charge motion observation, addressing core observation challenges in multimodal sensor fusion. The performance superiority of the proposed framework stems from the effective fusion of AEF and radar features. The initial feature selection (Section IV-B.1) confirmed the high discriminative power of the AEF features (SD and SE), and the final model's performance demonstrates that they provide unique information that complements the spatial context offered by the radar. The success of the model is, therefore, a result of this synergistic combination. Our key contributions include: 1) a new spatiotemporal calibration protocol resolving the critical 1 s-versus-6-min sampling misalignment between AEF sensor and weather radar, achieving 92.4% temporal alignment accuracy; 2) physics-constrained feature fusion with uncertainty-aware normalization, incorporating an AEF–REI interaction term to enhance observation robustness against nonstationary AEF signals; and 3) Bayesian-optimized localization reducing spatial error to 0.56 km (57.6% lower than threshold-based methods), validated against radar ground truth. The framework establishes a new remote sensing observation paradigm for extreme weather monitoring by integrating the 3DAEFA with radar systems, providing dynamic charge trajectory visualization previously unattainable with single-modal instruments, and advancing sensor fusion in meteorological observation through adaptive hyperparameter metrology. Future work will extend this observation framework to integrate lightning mapping arrays and micro-electromechanical system-based field sensors, enhancing real-time monitoring capabilities for operational early-warning systems.

Our framework establishes a novel paradigm for tracking charge motion by synergizing multimodal sensing. Future work will focus on overcoming the current limitations by: 1) deploying a multistation E-field sensor network to resolve the inherent ambiguity in charge inversion and better constrain complex charge structures; 2) integrating full 3-D radar volumetric data to correlate charge motion with reflectivity features at specific temperature altitudes; and 3) applying the framework to well-observed cases of isolated thunderstorm development to validate its performance under more controlled and simpler meteorological conditions. Furthermore, future validation will prioritize isolated thunderstorm cases to conclusively mitigate interpretation ambiguities associated with large-scale or multicell systems.

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